Kadhirezhilan D
192511228
ITA0612

CO2-AT3 - Debugging & Optimization Task

1. Perceptron model Fails to coverge

-> A perceptron is a machine learning model used to classify data into different classes. if the data is linearly seperable but the model does not converge, there may be errors in training

Possible causes:

i) Learning rate
ii) weight initialization
iii) Bias error
iv) update rule error

Debugging and optimization:

* check the weight and bias update formulas
* verify the input data and target labels
* Adjust the learning rate
* use a suitable stopping condition

CONCLUSION : correct implementation and Proper

2. Genetic Algorithm produces Poor solutions

2. Multilayer Perceptron Has Slow convergence

->A multilayer Perceptron (MLP) is a neural network with an input layer, hidden layers, and an output layer. Back Propagation is used to update weights and reduce errors.

Possible causes:

1) Vanishing gradients
2) Activation function
3) weight initialization
4) Learning rate

Debugging and optimization:

- check the loss value during training
- verify the backprogation calculations
- use suitable activation functions such as ReLU
- Normalize the input data
- use adaptive learning rates or optimizers such as Adam

CONCLUSION: Proper initialization, normalization, and suitable optimizers improve the speed and stability of training

3. Genetic Algorithm produces poor solutions

→ A Genetic Algorithm (GA) is an optimization technique inspired by natural selection. It uses selection, crossover, and mutation to find good solutions

Possible causes:

1) Incorrect chromosome encoding
2) Fitness function error
3) Premature covergence
4) Lack of diversity

Debugging and optimization

- check chromosome representation
- verify the fitness function
- Adjust the mutation rate
- control selection pressure
- use elitism to preserve the best solutions

CONCLUSION: Proper encoding, Population diversity, and suitable mutation and solution improve solution quality and convergence

4. Deep Neural Network may overfitting and high computational cost

→ A deep neural network uses multiple layers to learn complex patterns. Genetic programming can help generate or optimize solutions. However, the model may overfit and require high computational resources.

Possible causes:

1) complex model
2) over-Parameterization
3) Insufficient training data
4) overfitting

Debugging and optimization

- Compare training and validation performance
- use regularization to reduce overfitting
- Apply dropout to prevent excessive dependence on particular neurons
- use Pruning to remove unnecessary parameters

CONCLUSION: Regularization, dropout and prunning help improve generalization and reduce computational cost

5. Decision Tree has high Training Accuracy but Poor Test Performance

→ A decision tree is a machine learning model that makes predictions using decision rules. High training accuracy but poor performance on unseen data indicates overfitting

Possible causes:

1) overfitting
2) Deep tree structure
3) Noisy data
4) Poor feature selection

Debugging and optimization

- Compare training and testing accuracy
- use pruning to remove unnecessary branches
- Apply cross-validation to evaluate the model
- select relevant features
- Limit the maximum depth of the tree

CONCLUSION: Pruning, cross-validation and feature selection help the decision tree perform better on unseen data